

## **Introduction to Views in DBMS**

### **What is a View?**

- A view is a virtual table in a database.
- It is created from one or more tables.
- A view does not store data permanently.
- It displays data from existing tables.

### **Definition**

#### **View:**

A virtual table created using SQL query from one or more database tables.

### **Why Views are Used?**

1. Simplify complex queries
2. Improve security
3. Hide unnecessary data
4. Provide customized data access
5. Improve data abstraction

### **Important Points About Views**

6. View is not a real table.
7. Data comes from base tables.
8. Views can contain selected rows and columns.
9. Changes in base table reflect in view.
10. Views improve database security.

## Example of Base Table

### Student Table

**Student\_ID Name Marks Course**

101          Ravi    85      CSE

102          Anu     90      ECE

## Example of View

### SQL Query

```
CREATE VIEW Student_View AS  
SELECT Name, Marks  
FROM Student;
```

### Output of View

**Name    Marks**

Ravi     85

Anu      90

### Characteristics of Views

- 11.Views are virtual tables.
- 12.Views do not store actual data separately.
- 13.Views are created using SELECT statement.
- 14.Views can hide sensitive information.
- 15.Views can simplify repeated queries.

## **Types of Views**

### **1. Simple View**

#### **Definition**

- Created from one table only.

#### **Example**

```
CREATE VIEW Simple_View AS  
SELECT Name, Course  
FROM Student;
```

#### **Important Points**

- 16. Based on single table
- 17. Easier to manage
- 18. Supports insert and update

### **2. Complex View**

#### **Definition**

- Created from multiple tables.

#### **Example**

```
CREATE VIEW Complex_View AS  
SELECT Student.Name, Department.Dept_Name  
FROM Student  
JOIN Department  
ON Student.Dept_ID = Department.Dept_ID;
```

#### **Important Points**

- 19. Uses joins or functions
- 20. More complex than simple views

## **Operations on Views**

### **1. Create View**

```
CREATE VIEW View_Name AS  
SELECT column_name  
FROM table_name;
```

### **2. Display View**

```
SELECT * FROM View_Name;
```

### **3. Delete View**

```
DROP VIEW View_Name;
```

## **Advantages of Views**

- 21.Improves security
- 22.Simplifies complex queries
- 23.Hides unnecessary columns
- 24.Provides customized access
- 25.Reduces query complexity
- 26.Supports data abstraction
- 27.Improves data consistency
- 28.Easy report generation
- 29.Restricts unauthorized access
- 30.Reuses SQL queries

## **Disadvantages of Views**

- 31.Some views are slow
- 32.Complex views are difficult to update
- 33.Excessive views reduce performance
- 34.Maintenance becomes difficult in large systems

## Updating Views

### Definition

- Some views allow insert, update, and delete operations.

### Example

```
UPDATE Student_View  
SET Marks = 95  
WHERE Name='Ravi';
```

### Conditions for Updatable Views

- 35. View should use single table
- 36. Should not contain GROUP BY
- 37. Should not contain aggregate functions
- 38. Should not use DISTINCT

### Materialized View

#### Definition

- Stores data physically like a table.

#### Important Points

- 39. Faster query processing
- 40. Requires storage space

### Difference Between Table and View

| Table              | View              |
|--------------------|-------------------|
| Stores actual data | Virtual table     |
| Occupies storage   | Minimal storage   |
| Independent object | Depends on tables |
| Faster retrieval   | Sometimes slower  |

## **Real-Time Example**

### **College Database**

#### **Base Table**

- Student

#### **View**

- Display only Name and Marks to students.

### **Additional Important Points**

- 41.Views improve security.
- 42.Views are created using SQL.
- 43.Views can be simple or complex.
- 44.Data in views comes from tables.
- 45.Views support abstraction.
- 46.Views simplify database operations.
- 47.Changes in tables reflect in views.
- 48.Views help control access.
- 49.Views reduce repeated query writing.
- 50.Views are important in DBMS applications.

### **Short Definitions for Exams**

#### **View**

A virtual table created from one or more tables using SQL query.

#### **Simple View**

A view created from a single table.

#### **Complex View**

A view created from multiple tables.